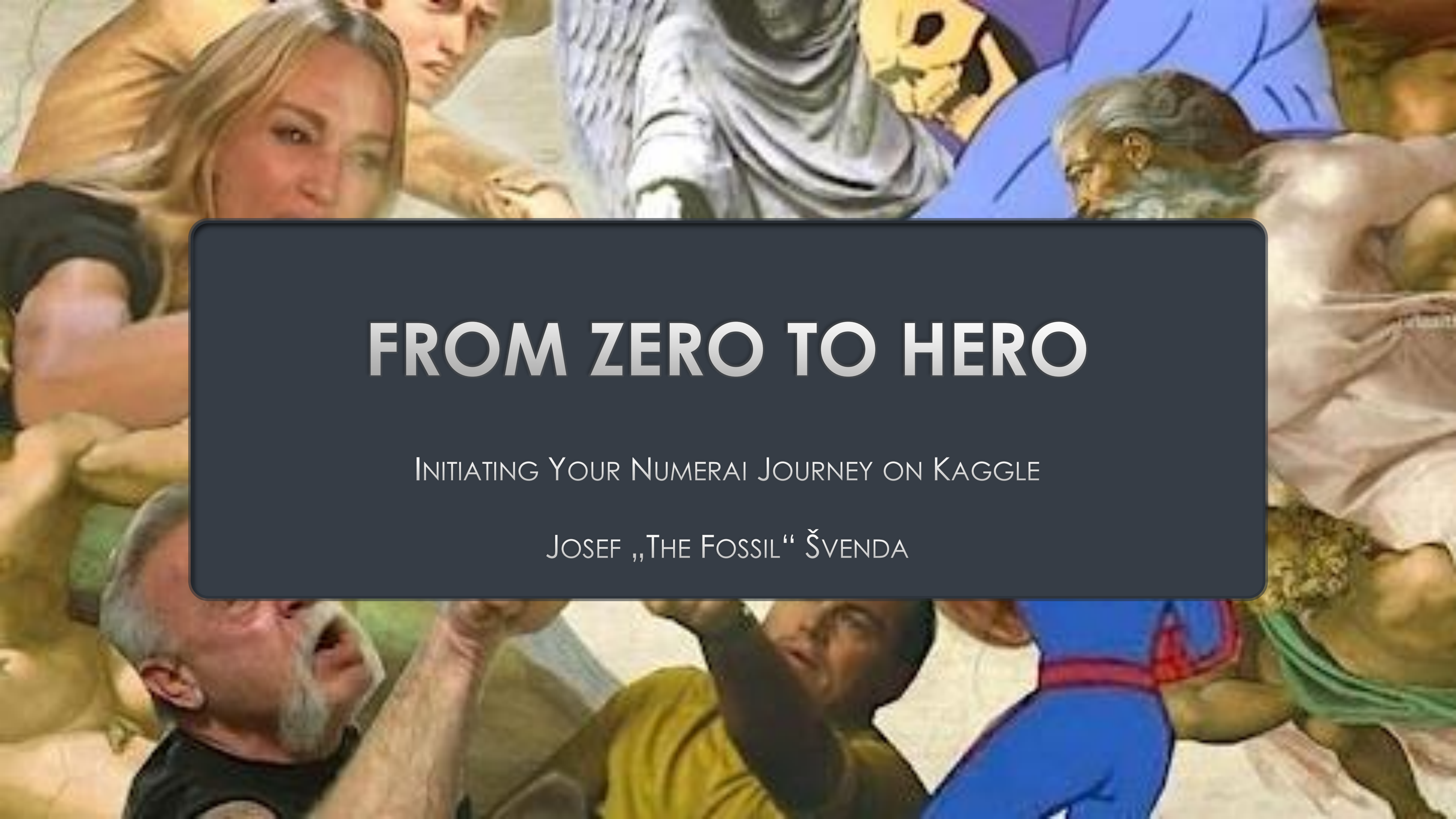


FROM ZERO TO HERO

Initiating Your Numerai Journey on Kaggle

Josef „The Fossil“ Švenda

AGENDA

What is Kaggle?

Numerai Data

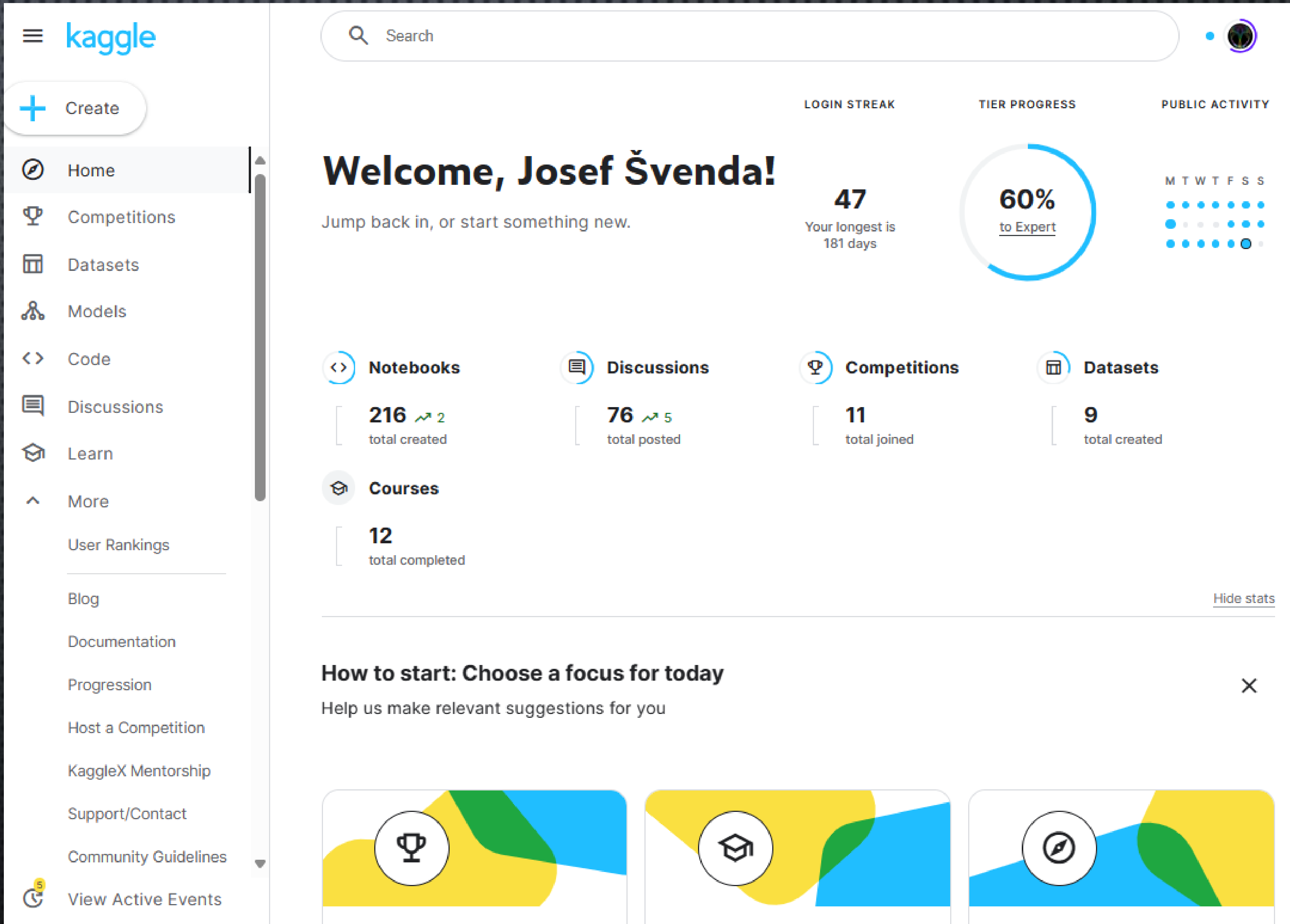
Numerai Example Notebooks

Numerai Automation Tips&Tricks

WHAT IS KAGGLE?

kaggle is a data science competition platform and online community of data scientists and machine learning practitioners under Google LLC.

As of October 2023, it has over 15 million users in 194 countries.



LIMITS OF FREE

WHAT YOU CAN DO IN KAGGLE

Limits

4 CPUs, 30GB RAM

GPU, 4 CPUs, 29 GB RAM, weekly max. 30 h

TPU, 96 CPUs, 330 GB RAM, weekly max. 20 h

Single run max. 12 hours CPU&GPU, 9h TPU

20GB disk space

Max 5 ntbks concurrently running

Chaining of notebooks

Datasets

Kaggle API

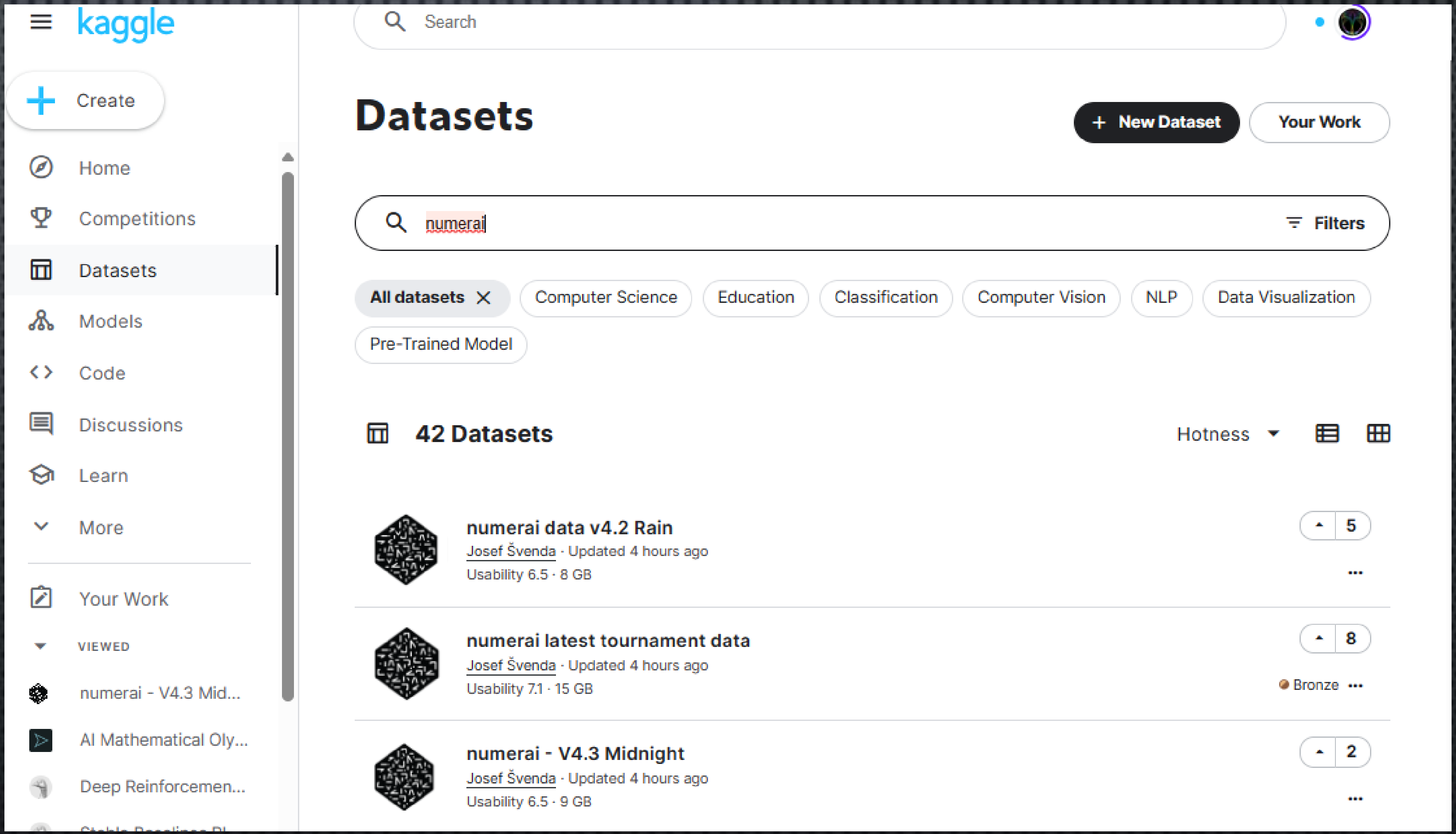
running Kaggle notebooks from outside

Enablers

NUMERAI DATASETS

Tournament data are weekly downloaded and stored as public Kaggle datasets:

- V4.3 Midnight
- V4.2 Rain
- V4 & V4.1

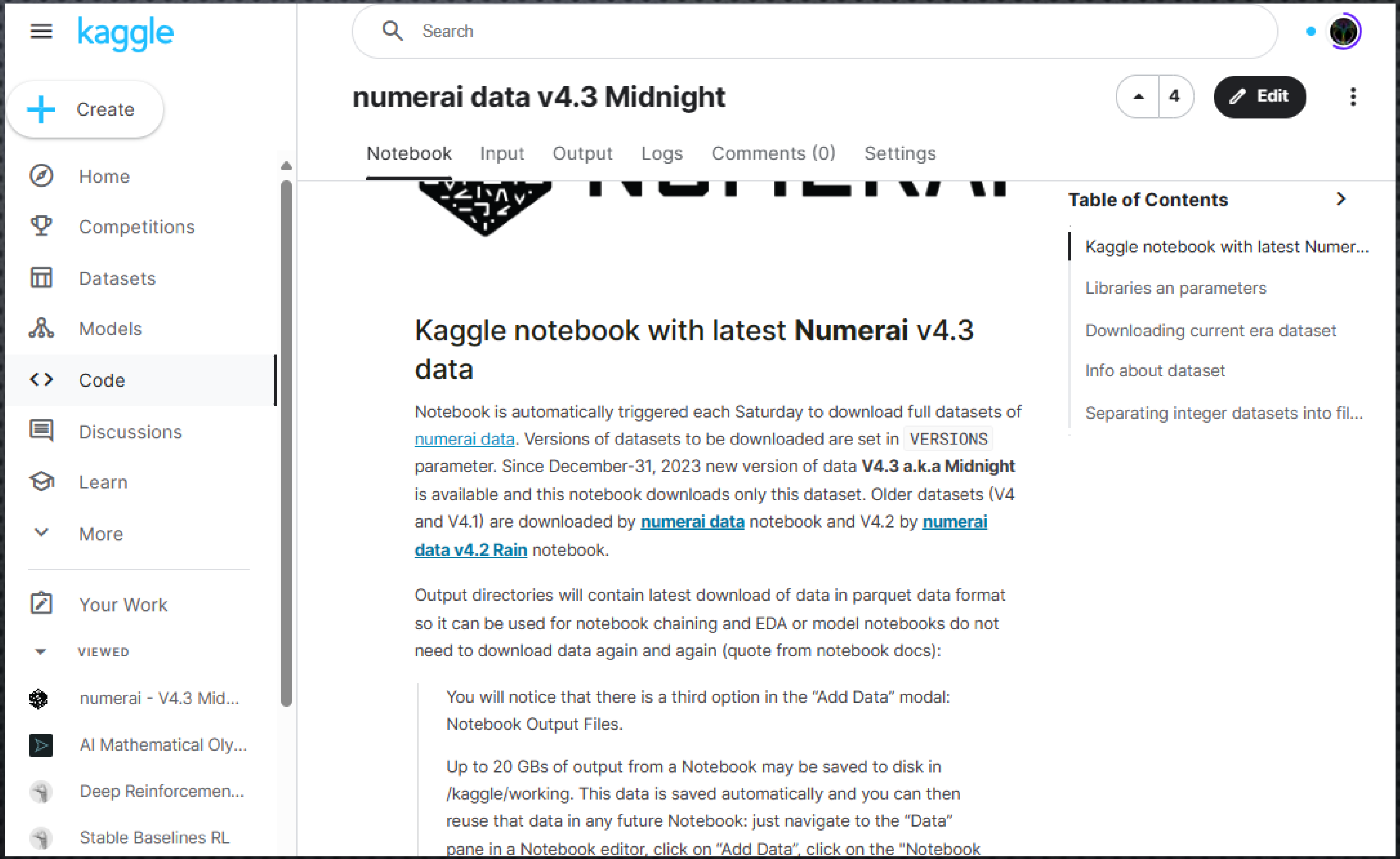


NUMERAI DATASETS

Each dataset has the producing notebook doing:

- Downloading current data
- „EDA“
- Producing smaller data files with „non-overlapping“ eras

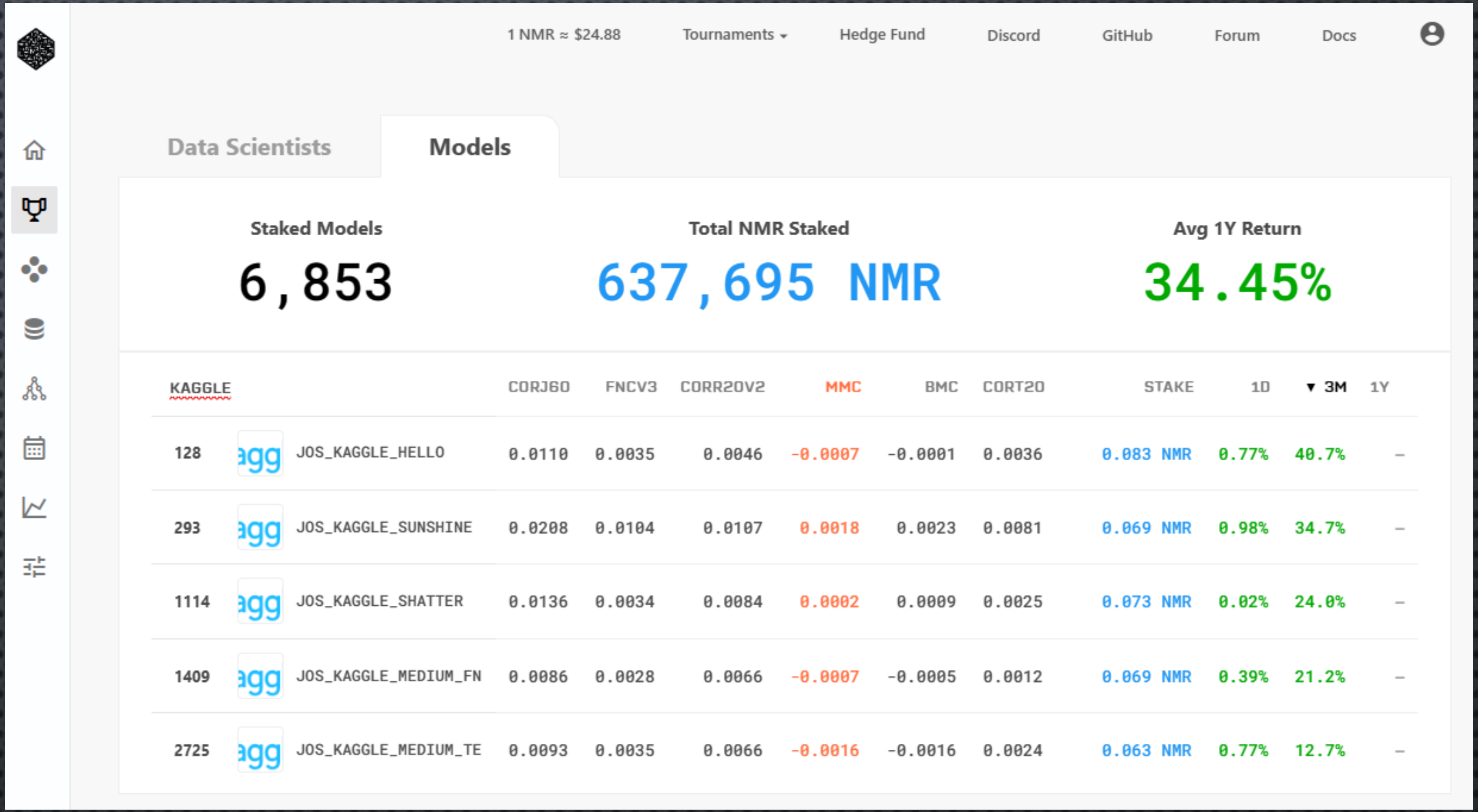
Either the dataset or the producing notebook can be used as input for other notebooks.



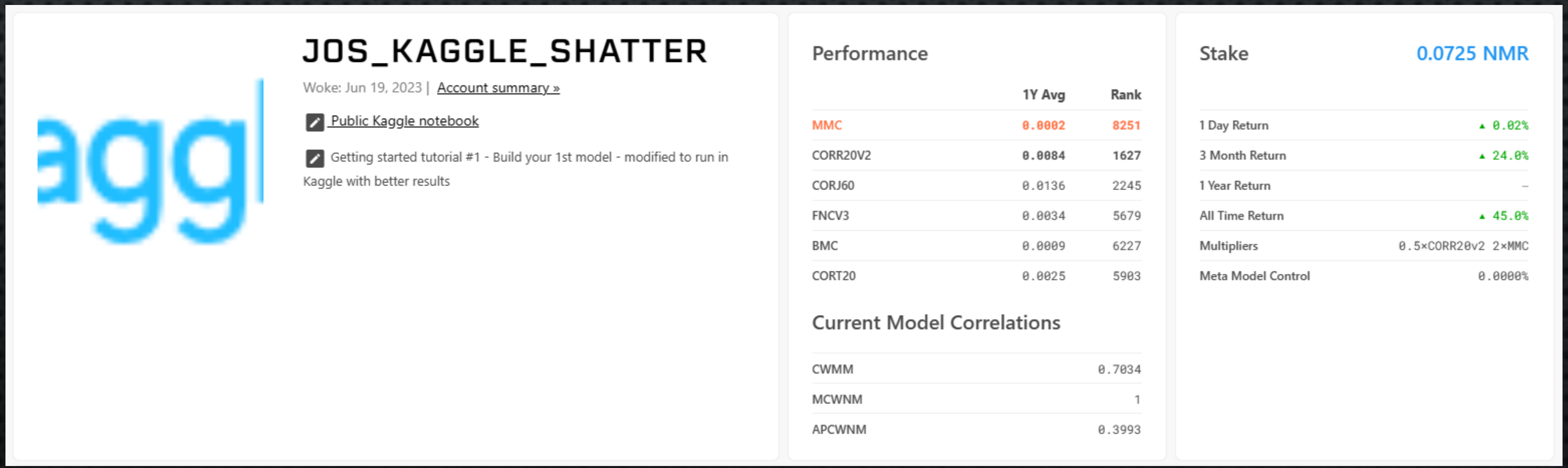
NUMERAI EXAMPLE NOTEBOOKS

There are 5 symbolically staked uploaded public models based on [Numerai tutorials](#):

- Hello Numerai
  - JOS_KAGGLE_HELLO – using small feature set
  - JOS_KAGGLE_SHATTER – using ShatteredX compact feature set
- Feature Neutralization
  - JOS_KAGGLE_MEDIUM_FN – using medium feature set
- Target Ensembles
  - JOS_KAGGLE_MEDIUM_TE – using medium feature set
- Older more complex Sunshine example
  - JOS_KAGGLE_SUNSHINE – using medium feature set



KAGGLE		CORJ60	FNCV3	CORR20V2	MMC	BMC	CORT20	STAKE	1D	▼ 3M	1Y
128	JOS_KAGGLE_HELLO	0.0110	0.0035	0.0046	-0.0007	-0.0001	0.0036	0.083 NMR	0.77%	40.7%	–
293	JOS_KAGGLE_SUNSHINE	0.0208	0.0104	0.0107	0.0018	0.0023	0.0081	0.069 NMR	0.98%	34.7%	–
1114	JOS_KAGGLE_SHATTER	0.0136	0.0034	0.0084	0.0002	0.0009	0.0025	0.073 NMR	0.02%	24.0%	–
1409	JOS_KAGGLE_MEDIUM_FN	0.0086	0.0028	0.0066	-0.0007	-0.0005	0.0012	0.069 NMR	0.39%	21.2%	–
2725	JOS_KAGGLE_MEDIUM_TE	0.0093	0.0035	0.0066	-0.0016	-0.0016	0.0024	0.063 NMR	0.77%	12.7%	–



Performance

	1Y Avg	Rank
MMC	0.0002	8251
CORR20V2	0.0084	1627
CORJ60	0.0136	2245
FNCV3	0.0034	5679
BMC	0.0009	6227
CORT20	0.0025	5903

Current Model Correlations

CWMM	0.7034
MCWNM	1
APCWNM	0.3993

Stake 0.0725 NMR

1 Day Return	▲ 0.02%
3 Month Return	▲ 24.0%
1 Year Return	–
All Time Return	▲ 45.0%
Multipliers	0.5×CORR20v2 2×MMC
Meta Model Control	0.0000%

JOS_KAGGLE_HELLO

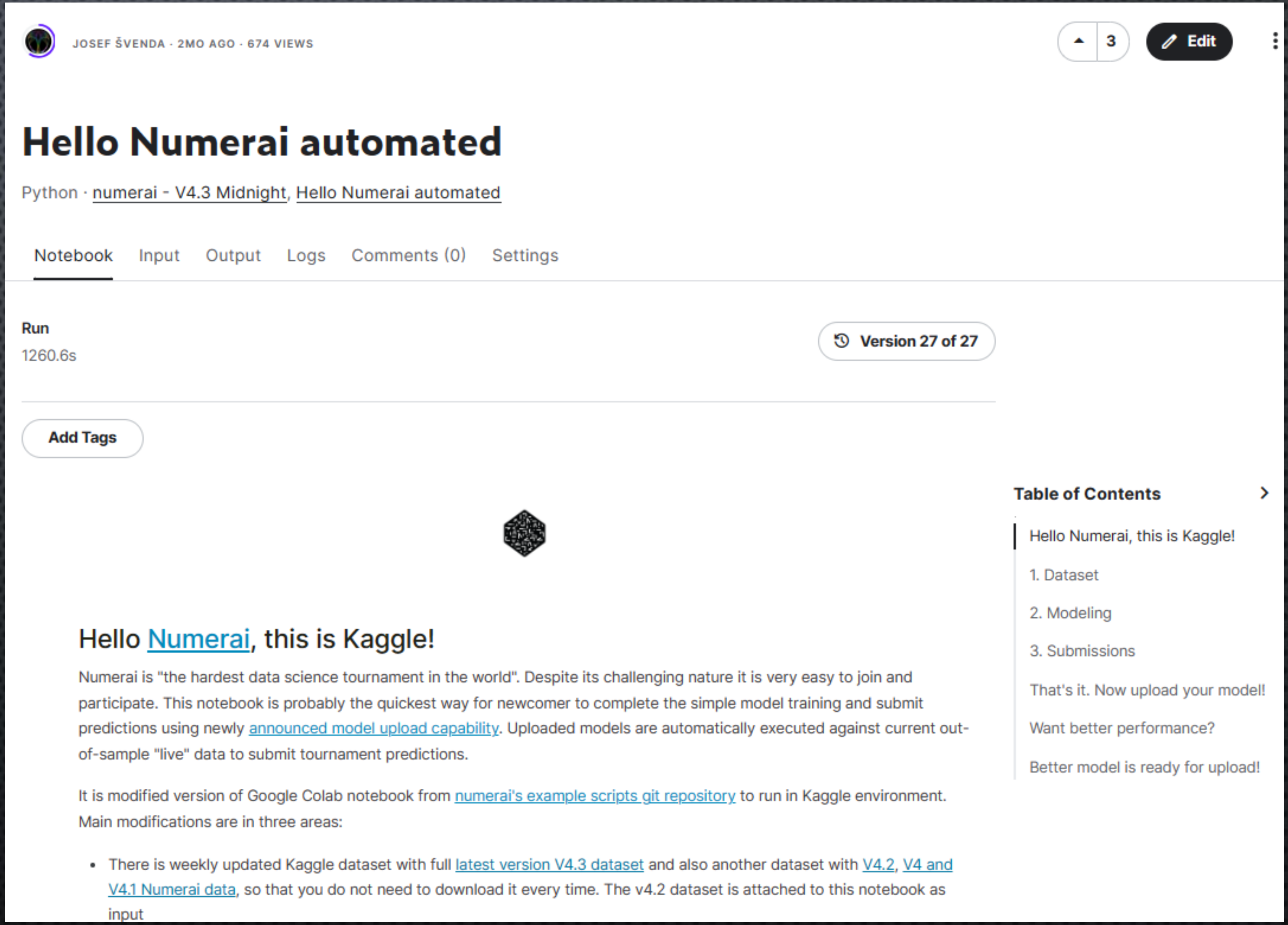
Hello Numerai automated (kaggle.com)

Typically consists from 3 main parts:

- Original example code slightly modified for Kaggle
  - Using Kaggle public dataset
  - Modifying Kaggle environment
- Model upload with NumerAPI
- Improving model performance usually by training on medium feature set

Modifications:

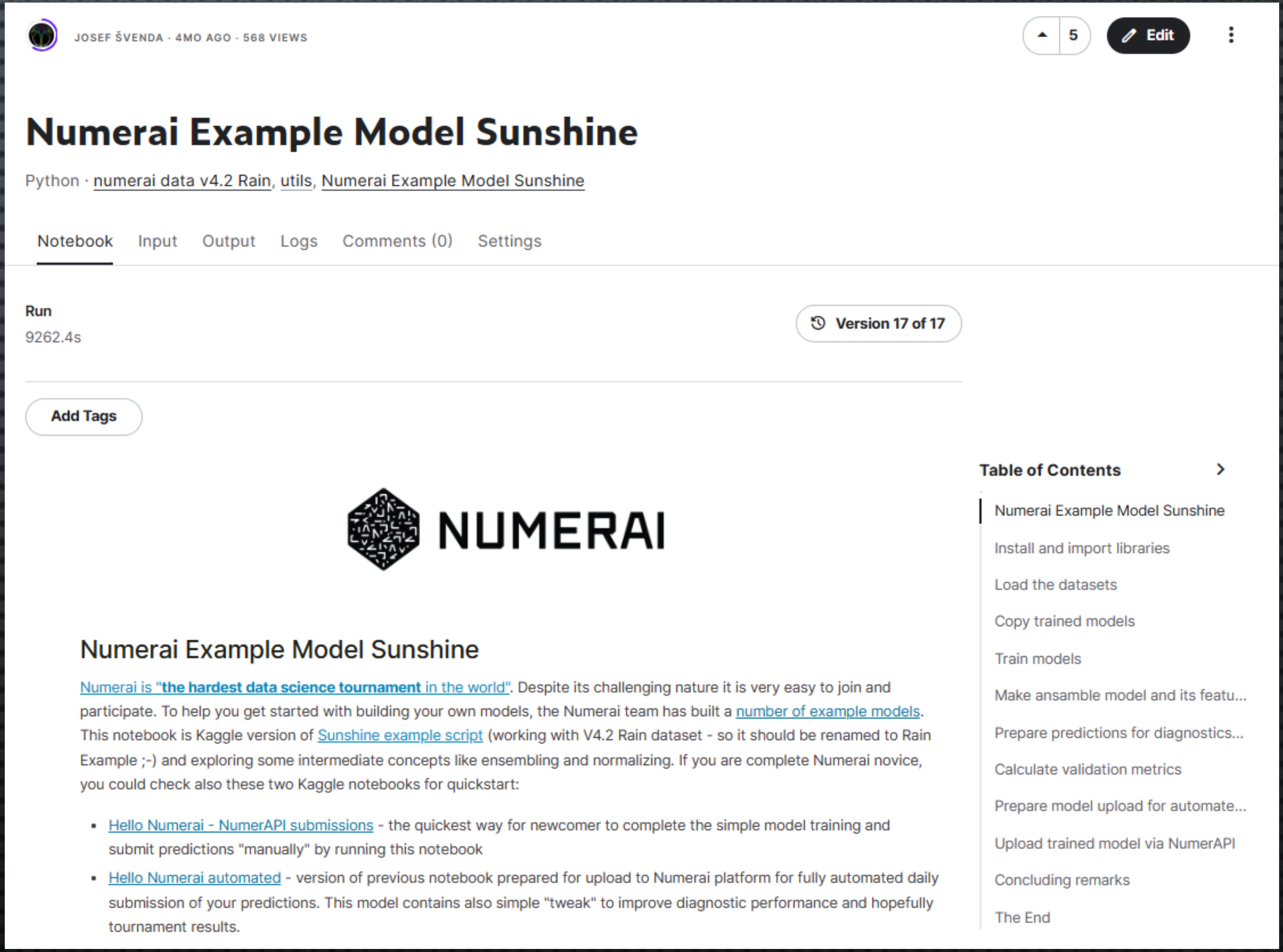
- Hello Numerai - NumerAPI submissions (kaggle.com) – submitting predictions via NumerAPI
- Hello Numerai automated on GPU (kaggle.com) – training on P100 GPU



JOS_KAGGLE_SUNSHINE

Numerai Example Model Sunshine (kaggle.com)

- Using old Numerai `utils` library for imports
- Both ensemble and neutralization
- Discussion of submission options
- Concluding remarks on various performance enhancement techniques



OTHER NOTEBOOK RESOURCES

Hyperparameter search:

- Numerai LightGBM Hyperparameter Search with Optuna (kaggle.com)
- Numerai XGBoost Hyperparameter Search with Optuna (kaggle.com)
- Numerai CatBoost Hyperparameter Search with Optuna (kaggle.com)

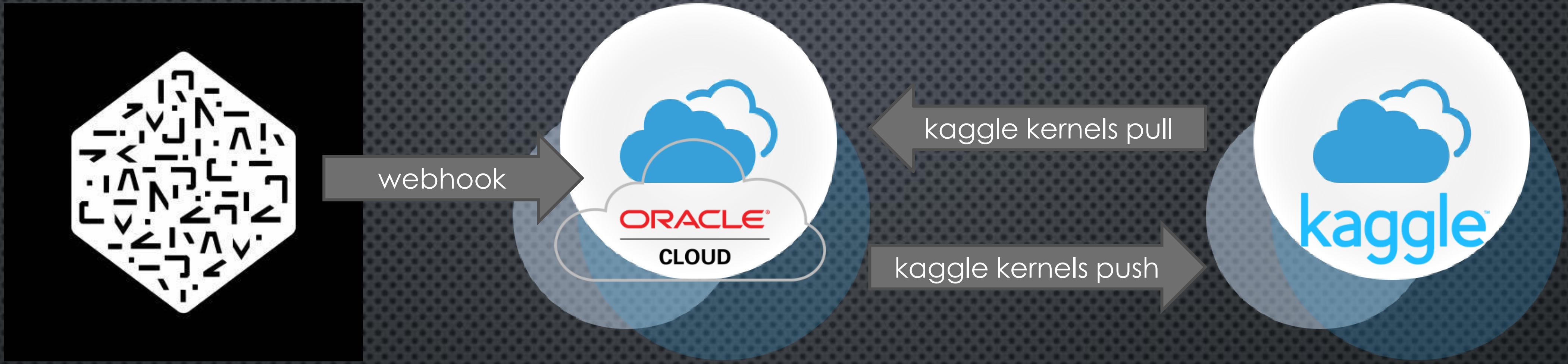
KATSU1110 – Numerai

- [numerai] MLP with KerasTuner Starter (kaggle.com)
- [Numerai] NN baseline with new massive data (kaggle.com)
- [numerai] factor analysis (kaggle.com)
- [numerai] PCA magic (kaggle.com)
- [NumeraiSignals] Starter for Beginners (kaggle.com)

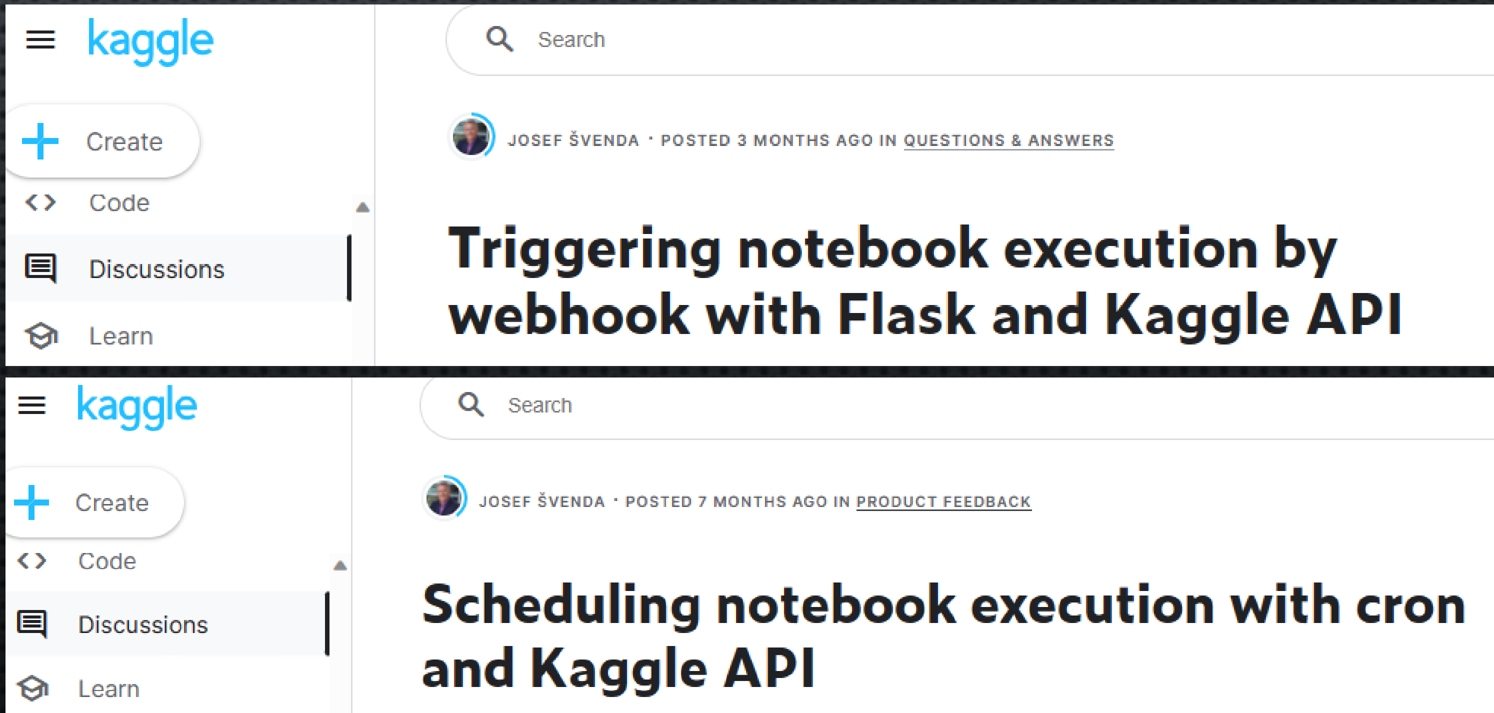
Carlo Lepelaars | Master | Kaggle of CrowdCent

- How to get started with Numerai (kaggle.com)
- Getting Started with Numerai Signals (kaggle.com)
- Numerai inference pipelines with NumerBlox (kaggle.com)

TRIGGERING RUN ON KAGGLE

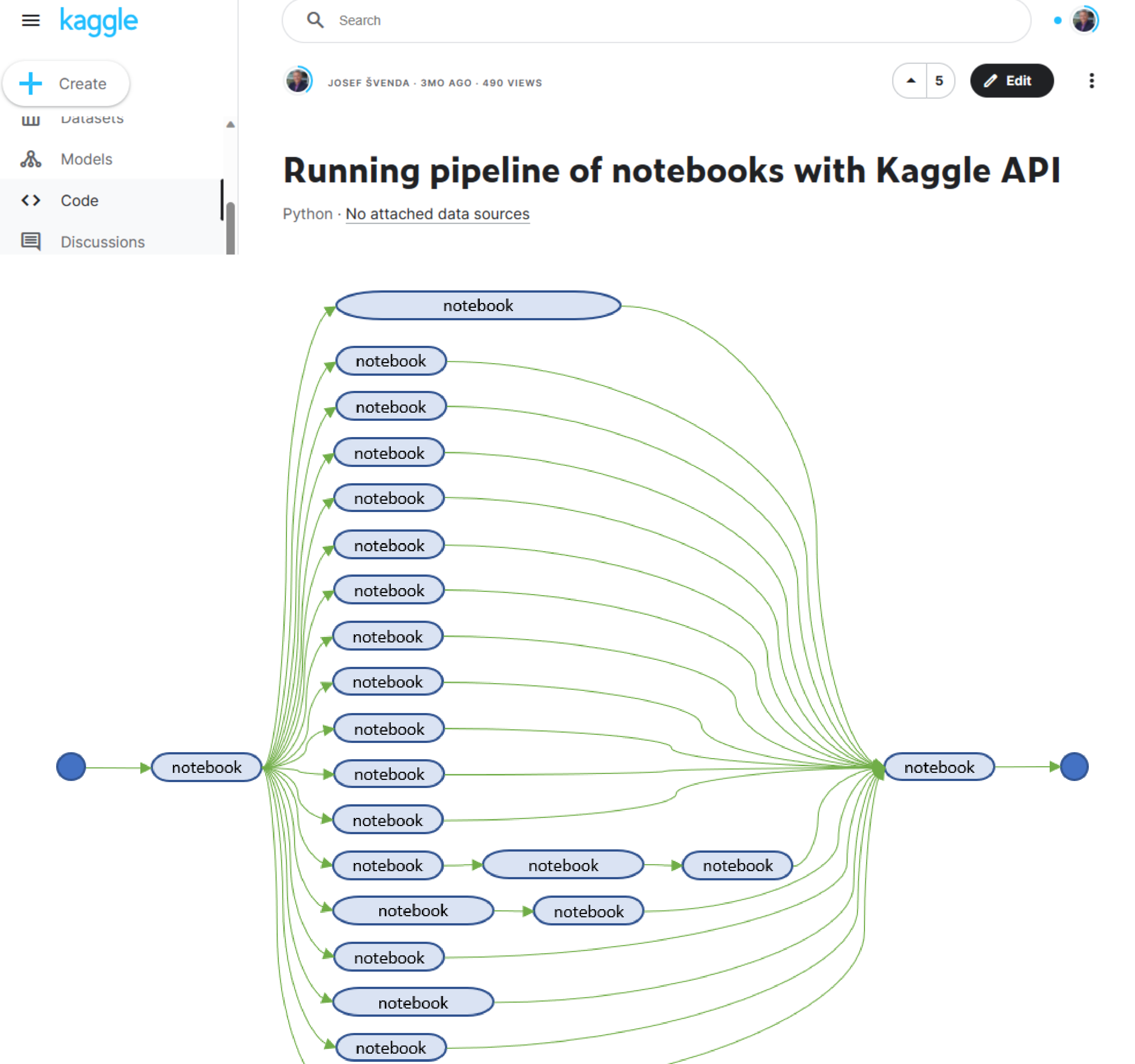


- Oracle Cloud, „Always FREE“, Ubuntu VM, 1 CPU with boot disk
- Python
- Kaggle API CLI
- Flask webserver



KAGGLE NOTEBOOK ORCHESTRATOR

- Running other notebooks
  - Sequentially
  - Parallel
- Must finish in 12 hours
- Kaggle API CLI
- Controlling python dict



BEST THINGS IN LIFE ARE FREE!!!

… BUT GOOD FINANCIAL REWARD IS ALWAYS APPRECIATED ;-)